

ADITYA GANDHAMAL

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RESEARCH AND PROFESSIONAL EXPERIENCE

Research Fellow

Sarvam AI

Bangalore

June 2025 — Present

- Building large-scale data generation pipelines for pre-training and SFT stages (150TB+, 260M+ images) for Sarvam Vision (3B-parameter VLM, 22 Indian languages), achieving SOTA on olmOCR-Bench and OmniDocBench.
- Converting curated datasets to WebDataset (WebDS) format for NVIDIA NeMo Framework-trainable pipelines; maintained internal Indic OCR benchmarks, designed evaluation metrics (RL-reward based test cases), and integrated frontier and open-source VLMs via vLLM-based distributed inference.
- Leading an applied research project for GoI (MoSPI)—a GIS + satellite + Gemma-4 + Grounding DINO pipeline to prioritize regions for the national Urban Frame Survey.

Pre-Doctoral Researcher

Artificial Intelligence and Robotics Laboratory — Indian Institute of Science

Bangalore

Aug 2024 — May 2025

- Conducted research on open-vocabulary learning, multi-modal alignment, and domain generalization in VLMs under Dr. Suresh Sundaram.
- Implemented techniques to adapt CLIP for open-vocabulary segmentation using Optimal Transport, Masked Image Modelling, and PeFT (LoRA, Prompt Learning); built multimodal benchmarks (EO-IR/sim-2-real/aerial-ground/BEV) for open-world settings.
- Developed CLIPcam (PyPI package for explainability) and delivered a talk on 3D Vision at a faculty development program.

Deep Learning and Image Processing Intern

GalaxyEye Space

Bangalore

April 2024 — July 2024

- Built a SAR-optical coregistration algorithm (Mutual Information + Powell Optimization), deployed aboard ISRO's POEM-4 payload, implemented code profiling and optimized runtime cutting it to 62% (40s → 15s).
- Diagnosed SAR distortion via Lucas-Kanade Optical Flow and built an image-warping autoencoder to model and rectify misalignment.
- Developed a SAR-EO fusion architecture (Knowledge Distillation + Contrastive Learning) as a PoC for the OptoSAR Satellite Launch—Mission Drishti.

Computer Vision Intern — AI for Healthcare

Indian Institute of Science

Bangalore

Nov 2023 — Feb 2024

- Worked on automation of drug delivery systems (Artificial Pancreas) under the guidance of Dr. Radhakant Padhi.
- Built a meal-detection pipeline (SAM + ResNet50 + fuzzy matching) for carbohydrate estimation, achieving 91% accuracy.
- Presented findings to a clinical panel from Ramaiah Hospital and Madras Diabetes Research Foundation.

Open Source Developer — Machine Learning

Unify (YC W23)

Remote

March 2023 — Oct 2023

- Contributed to Ivy, Unify's cross-framework ML transpilation framework.
- Created/reviewed PRs (functional APIs, transpilation utilities, bug fixes) across PyTorch, TensorFlow, JAX, NumPy backends.
- Recognized as a Top Contributor.

Research Intern — NLP

National Institute of Technology, Kurukshetra

Remote

March 2022 — Aug 2022

- Worked on Link Prediction on legal documents under the supervision of Dr. Sarika Jain.
- Built a Legal Knowledge Graph from Supreme Court filings, framing citation prediction and case similarity as link-prediction tasks.
- Benchmarked GraphSAGE, GAT, GCN vs. RGCN; GCN improved AUC ~40% (case similarity) and ~10% (citation prediction).

Project Intern — Computer Vision

Indian Army

Pune

March 2021 — Aug 2021

- Devised satellite imagery feature extraction techniques for road network identification, supporting urban planning and border surveillance, under Col. (Dr.) K. Joshil Raj.
- Curated annotated satellite imagery datasets and developed U-Net segmentation models for satellite road-network extraction, achieving 0.78/0.73 IoU on urban/border roads.
- Developed an interactive GUI enabling Army personnel to experiment with the models.

PUBLICATIONS

- Aniruddh Sikdar*, **Aditya Gandhamal***, Suresh Sundaram. “CoLoRA: Optimal Transport Meets LoRA for Multi-Domain Open-Vocabulary Segmentation” *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)* | [Project Page](#)
- **Aditya Gandhamal***, Aniruddh Sikdar*, Suresh Sundaram. “OV-COAST: Cost Aggregation with Optimal Transport for Open-Vocabulary Semantic Segmentation” *The 4th Workshop on Transformers for Vision (T4V) at CVPR 2025* | [Project Page](#)
- Aniruddh Sikdar*, **Aditya Gandhamal***, Suresh Sundaram. “AetherVision-Bench: An Open-Vocabulary RGB-Infrared Benchmark for Multi-Angle Segmentation across Aerial and Ground Perspectives” *Workshop on Foundation Models Meet Embodied Agents (FMEA) at CVPR 2025* | [Project Page](#)
- Sarvam AI — Vision Models Team. “Sarvam Vision: State-of-the-Art Vision-Language Model for Indic Document Intelligence” *Technical Blog, Sarvam AI* | [Technical Blog](#)
- Aniruddh Sikdar*, **Aditya Gandhamal***, Suresh Sundaram. “Tune CLIP as you Toggle (T-CAT): A Novel Toggle-Tune Based Prompt Learning Strategy for Open-Vocabulary Semantic Segmentation” *Under Review, Transactions on Machine Learning Research (TMLR)*
- **Aditya Gandhamal***, Aniruddh Sikdar*, Arya Kishor, Suresh Sundaram. “Pixel2Perspective: A Benchmark for Multi-Modal and Multi-View Generalization in Vision-Language Models” *Under Review, IEEE Transactions on Artificial Intelligence* | [Project Page](#)
- Manjunath D., Prajwal Gurunath, Sumanth Udupa, **Aditya Gandhamal**, Shrikar Madhu, Aniruddh Sikdar, Suresh Sundaram. “IndraEye: Infrared Electro-Optical UAV-based Perception Dataset for Robust Downstream Tasks” *arXiv preprint* | [arXiv](#)

EDUCATION

Savitribai Phule Pune University

Aug 2019 — June 2023

Bachelor of Engineering in Electronics and Telecommunication

- Relevant Coursework: Signal Processing, Control Systems, Digital Communication, Data Structures, Object Oriented Programming, Cloud Computing, Database Management Systems, Data Mining
- GPA: 8.3/10

PROJECTS AND OPEN-SOURCE CONTRIBUTIONS

CLIPcam `pip install clip-cam`



- Developed CLIPcam, an open-source Python package for visualizing prompt-image feature matching in ViT-based CLIP models; supports custom fine-tuned checkpoints, multiple CLIP backbones, and easy integration into VLM workflows for explainability.

Prompt Learning with CLIP for Open Vocabulary Segmentation



- Implemented prompt learning techniques for CAT-Seg (CVPR'24), integrating CLIP with Context Optimization (IJCV'22), Conditional Context Optimization (CVPR'22), and Textual-based Class-aware Prompting (CVPR'24); conducted ablation studies evaluating each variant against the MESS benchmark covering 20 multi-domain datasets.

Bias Mitigation for Fair Facial Attribute Classification (FAIR-Face)



- Fine-tuned ResNet-50 on CelebA for facial attribute classification while mitigating spurious correlations in “Bald” and “Male” attribute predictions; implemented adversarial training to reduce bias by 20%, evaluated using Equalized Odds.

Satellite Orbital Trajectory Forecasting (SatPO)



- Developed an LSTM-based time-series forecasting model trained on 18 days of telemetry data to predict satellite orbital trajectories for the next 7 days; validated performance through comparative analysis and trajectory visualization.

PyTorch-RL



- Added normalisation methods for reward function scaling in `RewardScaling` [\[PR\]](#); fixed redundant use of `step_mdp` in the data collector, rectifying incorrect state transitions during rollout [\[PR\]](#).

PyTorch-Vision



- Resolved a core validation bug where invalid strings were silently accepted as loss reduction modes; implemented `ValueError` handling across relevant loss functions and updated associated tests. [\[PR\]](#)

TALKS

LiDAR/Camera Fusion for 3D Object Detection | [\[slides\]](#)

Bangalore

Indian Institute of Science — Faculty Development Workshop on AI-Driven Autonomous Drones

Feb 11, 2025

- Explained 3D data representation techniques (point-based, voxel-based, pillar-based, frustum-based, BEV) and SOTA architectures across LiDAR-based methods (PointNet++, VoxelNet, PointPillars, SECOND), camera-based methods (FCOS3D, PETR, DETR3D), multimodal fusion architectures (MVX-Net, BEVFusion)
- Discussed real-world applications in autonomous driving and robotics, key challenges including sensor data sparsity and adverse weather robustness, and standard benchmarks — KITTI, Waymo, and nuScenes.

SKILLS

Languages: Python, Bash/Shell
Vision-Language Models: CLIP, Qwen, Gemma, PaliGemma, InternVL, PaddleOCR-VL
Training Infra & Fine-Tuning: Nvidia DGX Cloud Lepton, Yotta, SLURM, GCP Training Cluster, LoRA, Prompt Learning, Masked Image Modeling, Knowledge Distillation, Contrastive Learning, Context Optimization
Inference & Deployment: vLLM, Docker
Data & Pipelines: WebDataset (WebDS), Data Labelling and Deduplication, Pre-Training Data Curation, Synthetic Data Generation, YouTube Data API v3, Google Cloud Storage
Multi-Modal Sensor Data: RGB-SAR, RGB-IR, Multispectral Satellite Imagery (Cartosat)
Frameworks & Tools: PyTorch, HuggingFace Transformers, Detectron2, vLLM, TensorFlow, OpenCV, Git

ACHIEVEMENTS

- 2x Awardee of KIAC Fellowship ([Kotak-IISc AI-ML Centre](#))
- Algorithm developed during an internship at [GalaxEye Space](#) was successfully deployed and operational aboard the [GLX-SQ payload](#) launched as part of ISRO's [POEM-4 Mission](#).
- Contributed to building [Sarvam Vision](#), India's first sovereign Vision-Language Model, achieving state-of-the-art performance on Indic OCR and global document understanding benchmarks — Presented at the 2026 India AI Impact Summit, New Delhi.

VOLUNTEERING

AI/ML Co-Lead	Pune
<i>Google Developer Student Clubs</i>	<i>Aug 2021 — July 2022</i>
○ Organized and led Google Developer technology events, including hackathons and workshops. Conducted hands-on sessions on “Tensorflow” and “Git & GitHub”, reaching an attendance of 150 students overall.	
AI/ML Head	Pune
<i>ACM Student Chapter</i>	<i>Feb 2021 — July 2021</i>
○ Organized and delivered a webinar on "Python for Machine Learning" to around 100 students.	